

Codeplex Chronicle

YOUR CODEBASE, ALIVE

An AI-resident development environment for understanding, refactoring, and securing real production codebases, visualized as a living 3D city explored alongside five expert AI residents.

Product Requirements Document	v1.0
Team	Duopoly
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This document is the canonical product requirements specification for Codeplex Chronicle, submitted to Refactory Hackathon Round 03 evaluation.

1. Executive Summary

Codeplex Chronicle transforms real production codebases into **living 3D cities**. Files become buildings, folders become districts, errors become earthquakes you can feel. Five specialized AI residents live in iconic landmark buildings and serve five complementary product modes that address concrete engineering productivity pain.

The city is not the value proposition. The city is the **interface** and **memory hook**. Value lives in the five modes: Onboarding (new hire mental map in 30 minutes instead of two weeks), Sprint (spatial agile workspace with real-time PR-to-Building sync), Refactor (spec-driven refactor exploration in a safe sandbox before production), Activity (git backed engineering intelligence with ownership heatmaps), and Health (deterministic code diagnostic with one click backlog ticket creation).

Three Angle Differentiator

Other tools tell you what is in your code. Codeplex Chronicle tells you what your code promised versus what it actually delivered.

philosophical angle

Flat ticket lists don't show where work lives. Static code maps don't show what's planned. Codeplex Chronicle is one map of cities: your codebase, your sprint, and the gap between what you planned and what you shipped.

concrete angle

AI that hallucinates is expensive. A wrong refactor breaks production. Codeplex Chronicle gives AI room to explore without risk: every proposal and every refactor execution runs inside drafts/, not your production code. You see the blast radius with the ghost-to-solid animation, you read the diff calmly, you accept or you discard. Production only changes when you decide.

safety angle, closing punchline

At a Glance

Field	Value
Audience	Software engineers + engineering managers on GitHub-based workflows
Product modes	5 (Onboarding, Sprint, Refactor, Activity, Health)
AI residents	5 (Athena, Apollo, Argus, Clio, Hermes)
Mandatory stack	OpenSpec, LLMs (DeepSeek V4), Kubernetes, PostgreSQL
Build window	24 hours, hands-off operator mode
Submission	PRD, C4 diagram, repository link, slide deck

2. Problem Statement

Modern engineering teams operate inside codebases of 100K plus lines of code, with sprints tracked in one tool, code in another, and intent drift living silently in commit history. Four concrete pain points drive Codeplex Chronicle.

Pain 1. Engineer onboarding takes 2 plus weeks

A new engineer joining a 100K LOC codebase reads the README, opens the IDE, then loses two weeks navigating 200 plus files without a mental map. Existing tools (GitHub repo browser, IDE tree view, static code maps) are flat and provide no **active context**: who owns what, where the recent activity hotspots are, which file a new hire should touch first.

Pain 2. Managers context switch across three or more tools

Engineering managers move between GitHub (PR plus issue), Jira or Linear (sprint board plus velocity), custom dashboards (analytics plus burndown), and a wiki (spec plus decision log). Each tool has its own state model. Synchronisation is manual, and there is no spatial anchor that connects *where the work happens* with *where the code lives*.

Pain 3. Refactor proposals lack blast radius visibility

Engineers propose refactors as plain text inside a PR description or Jira ticket. Reviewers must mentally simulate impact: which files are affected, which dependencies break, where test coverage gaps lie. LLM-powered code assistants either write directly to production or to a branch without a visual review gate, so the blast radius only becomes visible after the diff already exists.

Pain 4. Spec to implementation drift is invisible

A sprint goal is defined in a milestone or issue. Implementation runs in a branch. The gap between what was specified and what shipped is not tracked. An issue closed six months ago, but the file keeps being edited. An OpenSpec change folder archived, but commits bypass the hook. These are silent drifts that only surface during manual audit.

Pain Validation

Pain	Confidence
Onboarding takes two plus weeks	High. Industry-standard observation across Stripe and Linear engineering blogs, plus team direct experience.
Manager context switch across three plus tools	High. Reinforced by team feedback on PR comment surfacing as a recurring bottleneck, plus past hackathon winner patterns favouring DevEx tooling.
Refactor blast radius gap	High. Drives the three angle pitch, central to product defensibility.
Spec to implementation drift	Medium. Five deterministic drift patterns validate the technical claim; the broader industry claim is partially validated by OpenSpec community discussion.

3. Solution Overview

The Big Picture

Codeplex Chronicle visualizes the entire codebase as a living 3D city. Files become buildings whose height scales with lines of code. Folders become districts. Imports become glowing roads. Test coverage becomes trees that fill well tested districts and thin out where coverage is missing. Ownership becomes the default colour of each building.

But the city is not just a visualization. It is an **operating canvas** that doubles as five distinct workspaces on a single shared surface.

Dual View, Dual Audience

Engineers work primarily in **City View** (the 3D city with PM overlay, refactor mode, ticket panel, and AI residents chat). Managers work primarily in **Dashboard View** (a flat 2D dashboard with velocity, burndown, milestone progress, contributor analytics, and a small embedded city preview in the corner). The view toggle lives in the top navigation. Engineers can zoom into the dashboard when they need velocity numbers quickly, and managers can zoom into the city for spatial context during one on one reviews.

Five Product Modes (Detail in Section 5)

Mode	Hero use case	What changes in the city
1. Onboarding	New hire mental map in 30 minutes	Camera flies through top three districts via Hermes guide
2. Sprint (HERO)	Spatial agile workspace	Scaffolding, cranes, banners, inspector NPCs, ticket panels
3. Refactor	Spec-driven refactor in a safe sandbox	Ghost buildings, OpenSpec change folder, drafts/ simulation
4. Activity	Git-backed engineering intelligence	Ownership heatmap, hotspot glow, timeline scrubber
5. Health	Code health diagnostic plus one click ticket	Glow buildings color-coded by severity, evidence panel

Five AI Residents (Detail in Section 6)

Resident	Landmark	Role
Athena	City Hall (Greek temple)	Refactor proposal author, OpenSpec change folder generator
Apollo	Hospital (cross-shape)	Code health diagnosis across five deterministic detectors
Argus	Police Station (camera tower)	Security triage with CVSS scoring and exploit pattern analysis
Clio	Library (book stack)	Git historian, sprint retrospective narrator, drift storyteller
Hermes	Tourist Info (glass beacon)	Codebase tour generator for onboarding and exploration

Core Safety Principle

AI in Codeplex Chronicle is a **narrator, navigator, summarizer, and proposal author with sandbox execution**. It never lands code on production autonomously. All insights are grounded in deterministic source: static analysis output, git and ticket metadata, OpenSpec proposals, or deterministic detectors. Refactor Mode executes simulation into a drafts/ folder, and production only changes when the user clicks Accept.

4. Target Users

Primary Persona A. Engineer Aldo

Full stack engineer with two to four years of experience, working in a team of five to fifteen at a startup or scale-up such as Tokopedia, Gojek, ruangguru, Sirclo, or Mekari. Daily tools are VSCode, GitHub, Slack, and one of Jira, Linear, or Trello.

Aldo is lost when onboarding to a large codebase, context switches constantly between IDE and ticket tool, and finds refactor proposals scary because there is no visual preview of the blast radius. **Success** for Aldo is being able to explain a codebase's mental model within 30 minutes of first contact, versus the two weeks that the current approach takes.

Primary Persona B. Engineering Manager Budi

Engineering manager with five to ten years of experience, leading eight to twenty engineers. Daily tools are GitHub web, Jira or Linear, custom dashboards, and one on one meeting tools.

Budi loses time on context switching between tools (velocity in Jira, code in GitHub, drift signals nowhere). He needs velocity reports quickly and a spatial anchor during one on ones with engineers. **Success** for Budi is generating a velocity plus drift summary in under one minute, versus the fifteen minutes that manual aggregation across three tools takes.

Secondary Persona. Refactory Hackathon Judge

An industry mentor or Refactory bootcamp alumnus evaluating the Round 03 submissions. The judge will *run the product themselves* on the demo laptop, per Refactory rules, and assess across five lenses: Innovation and Creativity, Impact and Relevance, Technical Execution, UX and Design, and Presentation and Communication.

Refactory criterion	Codeplex Chronicle angle
Innovation and Creativity	3D city + spec-drift detection + dual view + dual-folder OpenSpec strategy per panitia request
Impact and Relevance	Five real engineering productivity pains solved, plus Indonesian production code relevance via closing bonus slide
Technical Execution	OpenSpec first-class, K8s deploy, tree-sitter 11-language coverage, DeepSeek V4 multi-turn, r3f 60fps target
UX and Design	Awwwards-tier landing, cinematic intro, iconic landmark architecture, dual view, glassmorphism
Presentation and Communication	Three angle pitch flexibility, prepared Q and A defense cards, bahasa Indonesia primary with English code-switch

Selected User Stories

Eight hero stories from the full user story catalogue. Each story drives an acceptance criterion in Section 9.

Engineer Aldo

- **US-01.** As a new engineer, I want a 30-second tour of the codebase so I understand the three key districts I will touch most, plus their owners.
- **US-04.** As an engineer, I want the active sprint scope overlaid on the city so I understand where the work lives without switching to Jira.
- **US-06.** As an engineer, I want unresolved PR comments visible as annotations on a building so the bottleneck is immediately spatial.
- **US-08.** As an engineer, I want to type a plain language refactor intent, get a ghost building proposal plus an OpenSpec change folder, so the proposal is structured, not just a PR description.
- **US-11.** As an engineer, I want a clear Accept versus Done viewing decision after simulation, so I control when production changes.

Manager Budi

- **US-17.** As a manager, I want a Dashboard View with velocity, burndown, milestone progress, and contributor analytics by default, so I skip the context switch between tools.
- **US-19.** As a manager, I want a 60-second sprint retrospective flythrough with narration, so retro meetings move beyond flat slides.

Refactory Judge

- **US-23.** As a judge, I want a clear AI safety story (drafts/ sandbox, dual review gate, explicit Accept), so I am confident the product does not hallucinate production code.

5. Five Product Modes

Five complementary modes share the underlying 3D city and AI residents. A mode switcher lives in the top navigation. Each mode transforms how the city renders and what panels surface, while preserving the spatial anchor.

Mode 1. Onboarding

Hero use case

A new engineer joins the repository and needs a mental map fast. The outcome promise is a usable mental map in 30 minutes, not two weeks.

Flow

Click Hermes at the Tourist Info building. Type "give me a 30-second tour". The camera flies through the top three districts, chosen deterministically by ownership concentration and recent activity. Important buildings glow as the camera passes. Hermes narrates in the chat panel. The tour ends with a summary: starting point file, owner contact, and recommended next files.

Four tour variants

- "Give me a 30-second tour" for the generic new hire path.
- "Tour for sprint goal X" reads the milestone description.
- "Tour for feature Y" follows the dependency graph from the entry point.
- "Tour as @username" walks the districts that user owns or frequently contributes to.

Mode 2. Sprint Mode (HERO)

Hero use case

Active sprint scope is overlaid on the city. The mode transforms the city from a static codebase view into a live agile workspace, with no switch to a separate ticketing tool. This is the differentiator that judges remember most.

Fourteen PM concept visual mapping

Agile / PM concept	Visual
Active story or task	Scaffolding around the building
Open PR on file	Crane animating up and down during review
Backlog story	Blueprint pin floating over an empty slot
Done this sprint	Green halo glow, transient for 24 hours
Blocked story	Yellow tape and warning icon
Bug or issue type=bug	Smoke or visible crack on the building
Story points (size)	Size badge on the scaffolding (S/M/L/XL)
Sprint goal	Large banner on City Hall
Epic	District with highlighted border and central flag
Definition of Done	Checklist floating above the building
Code review (PR open)	Inspector NPC orbiting until approved
Dependency or blocker	Red glowing bridge between buildings
Refactor proposal	Ghost building with dashed outline
Spec drift	Crack visual with pattern A through E

PR-to-Building real-time sync

GitHub webhooks propagate PR state to building visuals in under five seconds. PR opened raises a crane. Review requested sends an inspector NPC. Approved triggers a green halo. Merged clears the scaffolding. An event store in PostgreSQL records every event for audit, cycle time, and lead time derivation.

PR comment surfacing

A new feature contributed from team observation: unresolved PR comments appear directly on the building as visual annotations. A PR pending with three unresolved comments shows a sticky note plus a badge counter. The exact visual approach (sticky note, floating bubble, or marker pin) is finalized at execution time by the Designer agent.

Dual view

Toggle to Dashboard View for a flat 2D manager-facing view with velocity chart, live burndown, milestone progress, and contributor analytics.

Mode 3. Refactor Mode (SAFETY-FIRST)

Hero use case

Spec-driven refactor exploration inside a sandbox, with two human review gates before any change touches production.

Nine-step flow

- **1. Intent input.** User types a plain language intent like "add 2FA to login".
- **2. Athena thinks.** Animation indicates processing. DeepSeek V4-Pro thinking mode runs deep analysis. V4-Flash handles simpler intents.
- **3. Ghost buildings appear.** Three transparent buildings render at the suggested location with dashed outline, signalling "proposed, not yet existing".
- **4. OpenSpec change folder auto-generates.** The side panel live streams proposal.md (ADDED requirements), design.md (approach and alternatives), and tasks.md (actionable checklist) into openspec/changes/<change-name>/.
- **5. Clio narrates trade-offs.** Optional context narration explains alternative approaches and their trade-offs.
- **6. Review Gate 1.** User reads the proposal in the side panel and clicks Run Simulation. This is not Apply, it is Simulate.
- **7. Simulation runs.** Backend coordinates DeepSeek V4 across three turns: test generation, implementation generation, diff serialization. Output streams to drafts/<simulation-id>/, never to production. The UI plays a ghost-to-solid animation frame by frame.
- **8. Simulation completes.** Ghost buildings solidify in simulation view. Side panel shows the diff plus stats (files changed, lines added, tests passed). Two buttons surface: Accept changes, or Done viewing simulation.
- **9. Review Gate 2.** Accept applies the diff to production. Done viewing keeps the simulation as a draft for later review or discard. Production code only changes here, on explicit user click.

Defensibility line

This is how we develop. We made it spatial. AI explores in drafts. You commit to production.

Mode 4. Activity Mode

Hero use case

Git-backed engineering intelligence. Surface what changed, who changed it, and where the hotspots are.

Visualisations

- Ownership concentration per district, heatmap colored by primary contributor.
- Contributor heatmap overlaying names on buildings owned.
- Hotspot intensity: buildings edited most recently glow brighter.
- Timeline scrubber for the last 30, 60, or 90 days.

Use case primary

- Daily standup. "Show me last 24 hours of activity."
- Sprint retro. A 60-second flythrough with Clio narration.

- Onboarding context. "Show me what @username has been working on."
- Performance review prep. "Summarize my contributions last quarter."

Mode 5. Health Mode

Hero use case

Code health diagnostic view with five deterministic detectors and a one click backlog ticket flow.

Five detectors monitored by Apollo

Detector	Mechanism
Hardcoded secrets	Regex plus entropy via gitleaks pattern
Outdated dependencies	Parse manifest files, cross-check OSV API for known CVE
Missing auth on protected routes	Parse decorators and middleware across TS/JS, Python, Go, Java, Rust
Unsafe SQL patterns	Detect raw string concatenation in queries, multi-language
Complex untested files	Cyclomatic complexity threshold plus test coverage stub

Flow

Switch to Health Mode. Buildings glow red (critical), orange (high), or yellow (medium) according to detector findings. Click Apollo at the Hospital building to open the findings panel. Click any finding for an evidence panel with file path, line number, static analysis output, and suggested mitigation. Click **Convert to Backlog Ticket** to POST to the GitHub API. A new issue is created with the evidence chain pre-filled, and an animation shows the issue flying from the affected building to the Backlog Office.

6. Five AI Residents

Five specialized AI residents live in iconic landmark buildings whose geometry signals their domain. Users either click the landmark or broadcast a question, and the relevant resident responds. The LLM provider is DeepSeek V4, with V4-Flash as primary and V4-Pro as fallback for complex reasoning.

Core safety philosophy. Every resident is a narrator, navigator, summarizer, or proposal author. None is an autonomous decision maker. All insights are grounded in deterministic source data.

Athena, City Hall (The Architect)

Landmark. Greek temple silhouette with pillar columns and pediment.

Domain. Refactor planning, spec authoring, dependency awareness, structural decision making.

Behavior. Athena is the OpenSpec proposal author in Refactor Mode. The user describes intent in plain language. Athena runs static analysis, identifies affected files and dependencies, proposes ghost buildings at suggested locations, and auto-generates the OpenSpec change folder with proposal.md, design.md, and tasks.md. Athena never lands code on production. Every proposal passes two human review gates.

Apollo, Hospital (The Doctor)

Landmark. Cross-shaped tower.

Domain. Code quality, complexity, test coverage, technical debt.

Behavior. Apollo runs five deterministic detectors (hardcoded secrets, outdated dependencies, missing auth on protected routes, unsafe SQL, complex untested files) and surfaces findings with an evidence chain: file path, line number, static analysis output. Apollo does not invent fixes. It surfaces evidence and suggests a direction. Findings can be routed to a one click GitHub ticket or escalated to Athena for refactor proposal.

Argus, Police Station (The Watcher)

Landmark. Surveillance camera tower geometry.

Domain. Security scanner and vulnerability triager. Overlaps with Apollo but specialises in security depth.

Behavior. Argus produces structured vulnerability reports with CVSS severity, exploit pattern (when documented in public CVE databases), and suggested mitigations referencing public advisories. Severe findings can be escalated to Athena for an OpenSpec refactor proposal rather than a simple issue.

Clio, Library (The Historian)

Landmark. Open book stack vertical.

Domain. Git archaeology, sprint retrospective narration, multi-mode prose narration including spec-drift storytelling.

Behavior. Clio narrates prose grounded entirely in deterministic metadata: timestamps, edit counts, reviewer identities. The LLM only composes the prose. All numbers come from git data. Example narration for a cracked building:

Building auth/oauth.ts is cracked under Pattern A. Issue #234 was closed eight months ago, but this file has been edited 12 times in the last three months. The implementation likely drifted from the original spec. Last reviewer: @hafiz. Last commit: two weeks ago.

Hermes, Tourist Info (The Guide)

Landmark. Glass beacon cube.

Domain. Codebase tour generation, narrative-driven exploration, onboarding guide.

Behavior. Hermes drives the camera through key districts based on a deterministic codebase analysis (ownership, dependency, recency). Hermes wraps structured data into engaging prose. The user can pause the tour mid-flythrough to explore manually, then resume to the next stop.

7. Spec-Drift Detection

Codeplex Chronicle visualizes the gap between code-as-implemented and code-as-specified. This is the philosophical differentiator: *other tools tell you what is in your code, we tell you what your code promised versus what it actually delivered.*

Three Building States

State	Definition	Detection
Solid (normal)	File exists, PR has touched it, PR linked to a merged issue, or change folder archived	Pure metadata join
Ghost	Path mentioned in proposal or issue body, milestone active, but path does not yet exist in the filesystem	Regex plus filesystem check
Cracked (retak)	Drift detected via five deterministic patterns	Pure deterministic detection

Five Crack Patterns

Pattern	Definition	Signal
A. Stale closed issue	Issue closed more than 6 months ago, file still being edited	Issue considered done but work continues
B. Closed without merge	Issue closed, no PR merged that touched the relevant file	Issue may have been dismissed but spec remains in tracker
C. Spec-implementation lag	Issue closed and file touched, but gap between closed timestamp and file last commit exceeds threshold	Implementation drifted from original spec
D. Reopened cycle	Issue reopened two or more times, or multiple closing PRs reverted	Spec keeps changing, building is unstable
E. OpenSpec drift	Commit touches a file referenced in an archived OpenSpec change, but commit message lacks the opsx: prefix	Hook bypass or merge from a branch without workflow guard

Clio as Narrator, Not Judge

Deterministic detectors flag the drift. Clio narrates the prose grounded in timestamps and metadata. The architectural distinction matters: AI is the storyteller, not the analyst. Patterns are extensible. New patterns can be added without changing the narration engine.

8. Architecture Overview

A full C4 model (Context, Container, Component, optional Code) is provided as a separate diagram artifact alongside this PRD. The summary below captures the system context and the key architectural decisions.

System Context

Codeplex Chronicle is a web SPA plus API that integrates with three external systems: GitHub (OAuth, REST API, webhooks), DeepSeek V4 inference, and PostgreSQL (pre-provisioned by Refactory).

External system	Purpose
GitHub	Source of truth for repositories, issues, milestones, PRs, OAuth identity, and webhook events
DeepSeek API (api.deepseek.com)	LLM inference for all five AI residents plus Refactor Mode multi-turn coordination. V4-Flash primary, V4-Pro fallback
PostgreSQL (Refactory managed)	Event store, cache, ticket state aggregation for cycle time and lead time observability

Container Diagram

Container	Tech
Frontend SPA	Next.js 16, React 19, Three.js 0.184, @react-three/fiber 9.6
Backend API	Python 3.12, FastAPI, Uvicorn
LLM Client Layer	OpenAI Python SDK pointing at api.deepseek.com
Database	PostgreSQL, Refactory-managed
OpenSpec Runtime	Fission-AI OpenSpec CLI, core profile
Ingress	NGINX, Refactory configured for HTTPS termination
Orchestration	Kubernetes, namespace duopoly, Refactory managed

Ten Key Architectural Decisions

#	Decision
AD-01	Monolith Next.js plus FastAPI, not microservice
AD-02	Treemap deterministic layout, not force-directed
AD-03	Raw instancedMesh with setMatrixAt for 200 to 300 buildings
AD-04	OpenSpec dual-folder strategy: openspec/ panitia-facing plus .agent-openspec/ internal workflow
AD-05	Visual frontend-first wave sequencing, backend deferred
AD-06	DeepSeek V4-Flash primary plus V4-Pro fallback, not Gemini
AD-07	GitHub Issues as backbone ticket (Sprint = Milestone, Story = Issue with label, PR = native)
AD-08	Refactor Mode writes to drafts/, dual review gate before production
AD-09	Five deterministic detectors for Health, AI as narrator only
AD-10	tree-sitter with 11-language grammar, lazy load per repo

9. Tech Stack

Refractory Mandatory Stack Compliance

Refractory Hackathon Round 03 mandates four technologies. Codeplex Chronicle uses all four naturally, not as bolt-on.

Required tech	Implementation in Codeplex Chronicle
OpenSpec	Fission-AI v1.0 stable, core profile. Dual-folder strategy per panitia request: openspec/ (panitia-facing primary spec) plus .agent-openspec/ (internal workflow output)
LLMs	DeepSeek V4-Flash primary plus V4-Pro fallback. OpenAI ChatCompletions API compatible. No OpenAI, no Anthropic, no Gemini at runtime
Kubernetes	Refractory pre-provisioned namespace duopoly with NGINX ingress, domain duopoly.hackathon.sev-2.com
PostgreSQL	Refractory-managed instance. Event store, cache layer, ticket state for cycle time and lead time observability

Frontend Stack

Layer	Stack	Version
Framework	Next.js with App Router	16
Language	TypeScript strict mode	5.x
UI runtime	React	19
3D core	Three.js (pinned, no auto-update)	0.184
3D React binding	@react-three/fiber	9.6
3D helpers	@react-three/drei	latest stable
Post-processing	@react-three/postprocessing (Bloom, DOF, Chromatic, Vignette, ACES tone mapping)	latest
Camera animation	GSAP	3.x
Styling	Tailwind CSS	3.x
Charts	Recharts or Chart.js	latest

Backend Stack

Layer	Stack	Notes
Language	Python	3.12 plus, async native
Web framework	FastAPI	Async, OpenAPI auto-doc, WebSocket for streaming
Static analysis	tree-sitter Python binding	11 language grammar: TS/JS, Python, Go, Java, C/C++, Rust, Ruby, PHP, Kotlin, Swift
Security detector	gitleaks pattern plus OSV API	Secrets plus CVE detection
GitHub backend client	PyGithub	Webhook receiver and issue creation
LLM client	OpenAI Python SDK with DeepSeek base_url	openai >=1.x
Database ORM	SQLAlchemy plus asyncpg	Async PostgreSQL
Migration	Alembic	Schema version control
Real-time	FastAPI WebSocket	Built in, for sync and streaming

Cost Posture

Hackathon runtime cost is bounded at approximately five dollars in total, covered by a team-subscribed DeepSeek account. GitHub API, Kubernetes, PostgreSQL, and the demo domain are all pre-provisioned at no cost by Refactory. The buffer is comfortable: at V4-Flash pricing of fourteen cents per million input tokens, five dollars represents roughly thirty million input tokens, far beyond demo and development needs.

10. Demo Flow

Each finalist team has ten minutes. The pitch core is two minutes punchy. The remaining eight minutes are Q and A plus technical depth.

Opening Hook (15 seconds)

A new engineer joins a 100K LOC codebase. Today she reads the README, opens VSCode, and gets lost in 200 files. Her manager asks for a velocity report, so he opens three different tools. We unified all of it in one city.

The Awwwards-tier landing page opens. Click Import a repository. The demo dataset loads. A five-second cinematic intro flies the camera in from low altitude. Lights turn on across the city as the residents wake up at their landmark buildings.

Mode 1: Onboarding (25 seconds)

Click Hermes at Tourist Info. Type "give me a 30-second tour". The camera flies through three districts as Hermes narrates. Ending: *this codebase has three districts you will touch most: auth, payment, api. Your starting point is auth/login.ts. Owner contact: @hafiz.*

Mode 2: Sprint Mode (45 seconds, HERO)

Switch to Sprint Mode. The city transforms. Scaffolding rises on buildings with active tickets. Cranes appear on buildings with open PRs. A banner reading *Sprint 14: Security hardening plus auth* appears on City Hall. Inspector NPCs orbit buildings under review. Click a scaffolded building. A ticket panel slides in: assignee, story points size:M, linked PR #45, status In Review. A sticky note on the building shows three unresolved comments.

Toggle to Dashboard View. Velocity chart, live burndown, milestone progress for the manager lens.

This is what we mean by spatial PM. Ticket lives where the work lives.

Mode 3: Refactor (30 seconds)

Switch to Refactor Mode. Type the intent: *add 2FA to login flow*. Athena thinks for two seconds. Three ghost buildings appear at `auth/oauth.ts`, `auth/totp.ts`, and `auth/middleware.ts`. The side panel live streams `openspec/changes/add-2fa/proposal.md`, `design.md`, and `tasks.md`.

Click Run Simulation. Tests are written first. The implementation lands incrementally. Ghost buildings solidify frame by frame. Simulation completes. Two buttons surface: Accept changes and Done viewing simulation. Click Done viewing.

AI explores in drafts. Production code only changes when you accept.

Mode 5: Health and One Click Ticket (25 seconds)

Switch to Health Mode on the NodeGoat demo dataset. Eight buildings glow red or orange. Click Apollo. *What is wrong?* Apollo responds: hardcoded API key at `config.ts` line 12, missing auth on `/admin` route, `payment.service.ts` cyclomatic complexity 52. Click Convert to Backlog Ticket on the first finding. An animation flies the issue from the affected building to the Backlog Office. Live confirmation appears in a GitHub tab side by side.

Closing Punchline (10 seconds)

Flat ticket lists don't show where work lives. Static code maps don't show what's planned. Codeplex Chronicle is one map of cities: your codebase, your sprint, and the gap between what you planned and what you shipped, all in one place. AI explores in drafts. You commit to production.

Closing slide: two pre-rendered bonus screenshots of *tokopedia/gripmock* and *gojek/courier-android* with the caption *works on real Indonesian production code*.

11. Q and A Defense Cards

Ten anticipated questions with prepared answers for the eight-minute Q and A window after the pitch.

#	Question	Answer
Q1	Scale to one million LOC?	Treemap deterministic plus LOD plus InstancedMesh plus frustum culling. Demo runs 300 files at 60fps. Theoretical scale 10K plus files with aggressive LOD. Production scale untested.
Q2	Private repo security?	Minimal GitHub OAuth scope: read:repo, read:issues, read:pull_requests, write:issues only. Token never leaves the user session. Code parsing is client-isolated, no cross-tenant leak.
Q3	How do AI residents avoid hallucination?	Every insight is grounded in static analysis output, git or ticket metadata, OpenSpec proposals, or deterministic detectors. AI is narrator, not analyst. Refactor Mode writes to drafts/, not production. User must explicitly click Accept.
Q4	Why 3D city, not 2D graph?	Spatial encoding bandwidth is higher than 2D. Verticality encodes LOC. District encodes folder. Glow encodes activity. Scaffolding encodes work in progress. Combined dimensions tell a story 2D cannot.
Q5	Compete with Jira or Linear?	Not compete, complement. We piggyback GitHub Issues as backbone. Teams keep their existing workflow. We add the spatial layer plus AI residents on top.
Q6	OpenSpec mandatory for users?	OpenSpec is first-class when the repo has openspec/. Without it, GitHub Issues are the informal spec source. Progressive degradation. For our own project we use dual-folder per panitia request: openspec/ for panitia-facing primary spec, .agent-openspec/ for internal workflow output.
Q7	What happens if DeepSeek API is down?	Defensive layer: semantic cache plus pre-recorded canned responses for the top ten demo questions. V4-Flash fallback if V4-Pro fails. Demo flow still works on canned responses in the worst case.
Q8	Why DeepSeek, not OpenAI or Anthropic?	DeepSeek V4 is 8 to 9 times cheaper per output token than GPT 5.5 or Claude Opus 4.7. OpenAI ChatCompletions API compatible, so the switch is drop in. Five dollars covers the entire hackathon with comfortable buffer.
Q9	What if drafts/ folder is lost?	Drafts are tracked by simulation-id UUID, persisted locally plus logged in the Postgres event store. Users can resume from a My Drafts panel. Orphan drafts are recoverable via simulation-id lookup.
Q10	DeepSeek is China-based. Data concerns?	Source code is sent to DeepSeek inference during Refactor Mode. The user is informed via the OAuth consent flow and privacy notice. For compliance-sensitive use cases, self-hosted DeepSeek V4-Flash on K8s is feasible (MIT licensed, weights on HuggingFace). MVP hackathon uses the hosted API.

12. Success Criteria

Hard Success Criteria (Must-Have for Ship)

#	Criterion
SC-01	Deployed product accessible at duopoly.hackathon.sev-2.com
SC-02	Five product modes operational: Onboarding, Sprint, Refactor, Activity, Health
SC-03	Five AI residents operational, each responding within five seconds of click
SC-04	Demo flow three times consecutive successful (pre-demo rehearsal gate)
SC-05	OpenSpec panitia-facing Folder A populated with feature spec
SC-06	C4 diagram formal and complete in deliverable bundle
SC-07	PRD plus slide deck submitted by Day 2 at 13:00 WIB
SC-08	Smoke test passes end to end without crash

Soft Success Criteria (Differentiation)

#	Criterion
SS-01	Visual quality bar baseline (glowing windows, fog, particles, post-processing pipeline, camera idle drift)
SS-02	Stretch Tier 1 shipped (cinematic intro, verticality skyscraper, iconic landmark architecture, Director mode auto-fly)
SS-03	Hybrid Write Layer 1 (Apollo to ticket) plus Layer 2 (Refactor simulation) both operational
SS-04	PR to Building real-time webhook latency below five seconds
SS-05	Closing slide bonus screenshots rendered

Acceptance Criteria Highlights

Hero acceptance criteria per mode, drawn from the full catalogue:

- **Onboarding.** Camera fly through three districts within 30 seconds. At least three buildings glow during the fly through. Hermes narration renders. Ending summary shows starting point file plus owner contact.
- **Sprint.** Scaffolding on every active-ticket building. Crane on every open-PR building. PR opened webhook propagates within five seconds. Ticket panel opens in under 300 ms on click.
- **Refactor.** Within ten seconds of intent, three ghost buildings render with dashed outline. OpenSpec change folder content streams live. Run Simulation triggers DeepSeek multi-turn. Ghost-to-solid animation plays frame by frame. Two buttons (Accept versus Done viewing) surface on completion.
- **Health.** Five detectors run on NodeGoat, all five fire. Buildings glow with color-coded severity. Convert to Backlog Ticket creates a GitHub issue with the evidence chain pre-filled.

13. Team and Closing

Tim Duopoly

Codeplex Chronicle is built by a two-person team operating under a **hands-off model**: the operators relay handoff between AI agents, approve decisions, and ferry knowledge between contexts. Code execution is fully delegated to Claude Code workers, including Kubernetes deploy and the slide deck generation pipeline.

Member	Role
Ghaisan Khoirul Badruzaman	Workflow orchestration, agentic pipeline design, PRD authoring, pitch lead
Hafiz Fauzan Syafrudin	Operator partner on sleep-cycle handoff, Kubernetes deploy familiarization, slide deck author, Day 2 final presentation

Submission Bundle

- **PRD** (this document, plus the Markdown agent-consumed version)
- **C4 diagram** formal and detailed, generated by Claude Code agents
- **Repository link**: github.com/Finerium/codeplexRefactory
- **Slide deck**: authored by Hafiz on Day 2

Closing

Codeplex Chronicle treats the codebase as a place. Engineers walk through it. Managers fly over it. AI residents speak about it. Refactor proposals visit it in ghost form before they settle as solid buildings. Spec drift cracks the walls until someone patches the breach.

Five product modes. Five AI residents. One city. All of it grounded in deterministic source, OpenSpec first-class, and DeepSeek V4 inference. Production code only changes when the user decides.

Your codebase, alive.

Tim Duopoly · Refactory Hackathon Round 03 · Telkom University Bandung · 12 to 13 May 2026